

MINOR PROJECT REPORT

on

"GLITCHLABS: Multiplayer Debugging and Code Reasoning Platform"

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Submitted By:

Name of Student 1 (Registration No.)

Name of Student 2 (Registration No.)

Name of Student 3 (Registration No.)

Under the Supervision of

Dr./Mr./Ms. Guide Name

(Designation)

Department of Computer Science and Engineering

JECRC University, Jaipur

Session: 2025–26

DECLARATION

I hereby declare that the project work entitled **“GLITCHLABS: Multiplayer Debugging and Code Reasoning Platform”** is an original work carried out under the supervision of the project guide in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at JECRC University, Jaipur. This work has not been submitted previously for any degree or diploma at any institution or university.

Signature of Student

CERTIFICATE

This is to certify that the Minor Project titled **“GLITCHLABS: Multiplayer Debugging and Code Reasoning Platform”** has been successfully completed by the students of B.Tech (CSE), JECRC University, Jaipur, under the supervision of the undersigned during the academic session 2025–26.

Project Guide Signature

Head, Department of CSE

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project guide, faculty members, and the Department of Computer Science and Engineering for their valuable guidance and continuous support throughout the development of this project. We also thank our peers and friends for their suggestions, encouragement, and feedback during the implementation and testing phases of GLITCHLABS.

ABSTRACT

*GLITCHLABS is a multiplayer debugging and code reasoning platform designed to help students and programmers improve their debugging and analytical coding skills through interactive gameplay. Traditional coding platforms focus mainly on writing code, whereas debugging skills are often neglected despite being essential in software development. The proposed system introduces a structured debugging environment where players collaborate or compete in real-time coding sessions to identify, analyze, and fix bugs in programs. The platform uses technologies such as FastAPI, React, WebSockets, Python, and modern frontend frameworks to create a real-time multiplayer experience. The project incorporates debugging workflows such as hypothesis generation, code tracing, bug fixing, and result analysis. Different levels of difficulty and multiple programming language support make the platform engaging and educational. The outcome of the project is an innovative educational platform that transforms debugging practice into an interactive and competitive learning experience, helping users strengthen problem-solving abilities, logical reasoning, and software debugging expertise. **Keywords:** Debugging, Multiplayer Platform, FastAPI, React, WebSockets, Competitive Programming, Code Analysis*

TABLE OF CONTENTS

Chapter	Title	Page
1	Introduction	1
2	Literature Review	5
3	System Design	10
4	Implementation	18
5	Testing & Results	26
6	Conclusion & Future Work	32

CHAPTER 1: INTRODUCTION

GLITCHLABS is an educational multiplayer debugging platform that focuses on improving debugging and analytical reasoning skills among programmers. Most coding platforms emphasize problem solving and algorithm implementation but provide limited focus on understanding program behavior and debugging. GLITCHLABS addresses this issue by introducing a game-based debugging environment. The system allows users to join multiplayer rooms, participate in debugging rounds, identify logical and syntactical errors, and compete based on accuracy and speed. The project aims to make debugging more engaging while improving real-world programming skills.

CHAPTER 2: LITERATURE REVIEW

Various online coding platforms such as LeetCode, HackerRank, and Codeforces focus on coding challenges and competitive programming. However, debugging-oriented platforms remain limited. Existing educational systems often lack collaborative debugging and real-time gameplay elements. GLITCHLABS differentiates itself by integrating debugging workflows with multiplayer interaction, gamification, and real-time synchronization technologies.

CHAPTER 3: SYSTEM DESIGN

The system architecture of GLITCHLABS consists of a React frontend, FastAPI backend, WebSocket communication layer, and a database system for storing users, problems, and results. The frontend provides an interactive user interface for room creation, gameplay, and debugging visualization. The backend handles authentication, game logic, matchmaking, and synchronization between users.

CHAPTER 4: IMPLEMENTATION

The implementation phase involved integrating FastAPI with WebSockets for real-time communication and React for frontend rendering. The system supports

multiplayer rooms, code editors, timers, hypothesis submission, and debugging workflows. Python was used for backend services while JavaScript and React were used for the frontend. Additional libraries and APIs were integrated for enhanced gameplay and debugging analysis.

CHAPTER 5: TESTING & RESULTS

The project was tested for functionality, usability, and performance. Multiple test cases were created to validate room synchronization, debugging flow, user interaction, and response times. Results demonstrated that the platform effectively supports real-time debugging gameplay with low latency and accurate synchronization across connected players.

CHAPTER 6: CONCLUSION & FUTURE WORK

GLITCHLABS successfully introduces a unique approach to debugging education by combining multiplayer interaction with structured debugging workflows. The platform enhances logical reasoning and debugging skills while maintaining user engagement through gamification. Future improvements include AI-generated debugging problems, advanced analytics dashboards, ranking systems, and deployment for large-scale multiplayer support.

BIBLIOGRAPHY / REFERENCES

1. FastAPI Documentation – <https://fastapi.tiangolo.com/>
2. React Documentation – <https://react.dev/>
3. WebSocket Protocol RFC 6455
4. Python Software Foundation Documentation
5. Research papers related to debugging education and gamification